Liquidity Monitoring and Measurement

Stress Testing

Liquidity stress testing is performed for each of Citi's major entities, operating subsidiaries and/or countries. Stress testing and scenario analyses are intended to quantify the potential impact of an adverse liquidity event on the balance sheet and liquidity position, and to identify viable funding alternatives that can be utilized. These scenarios include assumptions about significant changes in key funding sources, market triggers (such as credit ratings), potential uses of funding and geopolitical and macroeconomic conditions. These conditions include expected and stressed market conditions as well as Company-specific events.

Liquidity stress tests are conducted to ascertain potential mismatches between liquidity sources and uses over a variety of time horizons and over different stressed conditions. Liquidity limits are set accordingly. To monitor the liquidity of an entity, these stress tests and potential mismatches are calculated with varying frequencies, with several tests performed daily.

Given the range of potential stresses, Citi maintains contingency funding plans on a consolidated basis and for individual entities. These plans specify a wide range of readily available actions for a variety of adverse market conditions or idiosyncratic stresses.

Short-Term Liquidity Measurement: Liquidity Coverage Ratio (LCR)

In addition to internal liquidity stress metrics that Citi has developed for a 30-day stress scenario, Citi also monitors its liquidity by reference to the LCR, as calculated pursuant to the U.S. LCR rules.

Generally, the LCR is designed to ensure that banks maintain an adequate level of HQLA to meet liquidity needs under an acute 30-day stress scenario. The LCR is calculated by dividing HQLA by estimated net outflows over a stressed 30-day period, with the net outflows determined by applying prescribed outflow factors to various categories of liabilities, such as deposits, unsecured and secured wholesale borrowings, unused lending commitments and derivatives-related exposures, partially offset by inflows from assets maturing within 30 days. Banks are required to calculate an add-on to address potential maturity mismatches between contractual cash outflows and inflows within the 30-day period in determining the total amount of net outflows. The minimum LCR requirement is 100%, effective January 2017.

Pursuant to the Federal Reserve Board's final rule regarding LCR disclosures, effective April 1, 2017, Citi began to disclose LCR in the prescribed format.

The table below sets forth the components of Citi's LCR calculation and HQLA in excess of net outflows for the periods indicated:

In billions of dollars	**Dec. 31, 2017**	Sept. 30, 2017	Dec. 31, 2016
HQLA	**$ 446.4**	$ 448.6	$ 403.7
Net outflows	**364.3**	365.1	332.5
LCR	**123%**	123%	121%
HQLA in excess of net outflows	**$ 82.1**	$ 83.5	$ 71.3

Note: Amounts set forth in the table above are presented on an average basis.

As set forth in the table above, Citi's LCR increased year-over-year, as the increase in the HQLA (as discussed above) more than offset an increase in modeled net outflows. The increase in modeled net outflows was primarily driven by changes in assumptions, including changes in methodology to better align Citi's outflow assumptions with those embedded in its resolution planning. Sequentially, Citi's LCR remained unchanged.

Long-Term Liquidity Measurement: Net Stable Funding Ratio (NSFR)

In 2016, the Federal Reserve Board, the FDIC and the OCC issued a proposed rule to implement the Basel III NSFR requirement.

The U.S.-proposed NSFR is largely consistent with the Basel Committee's final NSFR rules. In general, the NSFR assesses the availability of a bank's stable funding against a required level. A bank's available stable funding would include portions of equity, deposits and long-term debt, while its required stable funding would be based on the liquidity characteristics of its assets, derivatives and commitments. Prescribed factors would be required to be applied to the various categories of asset and liabilities classes. The ratio of available stable funding to required stable funding would be required to be greater than 100%. While Citi believes that it is compliant with the proposed U.S. NSFR rules as of December 31, 2017, it will need to evaluate a final version of the rules, which are expected to be released during 2018. Citi expects that the NSFR final rules implementation period will be communicated along with the final version of the rules.